

We use silicon waveguides to carry light signals on chips. Optical modulators in these circuits encode data by changing light's phase or amplitude, we need modulators because lasers alone cannot carry information at high speed and with high fidelity — modulators encode electronic data onto light for communication, sensing, and computation.

In optical transmitters, modulation is generally performed either by directly modulating the laser, i.e., by turning on and off the laser, or by modulating the light out of the laser using an external modulator. Direct modulation can be more compact, simpler, and could also be cost effective. External modulation, however, has reached higher speeds, has larger modulation depth, and introduces less distortions [5].

The modulation of the phase and/or amplitude of optical signals in silicon waveguides can be achieved by changing the refractive index and optical absorption of silicon.

Electronics concepts relating modulators with silicone photonics

- Knowing that Silicon is a semiconductor, doping creates p-type (holes) and n-type (electrons) regions. A p–n junction diode in silicon can be forward- or reverse-biased to inject or remove free carriers. Free electrons/holes in the waveguide change its refractive index (and absorption) via the plasma dispersion effect (where the concentration of carriers is varied either by injecting or removing carriers from the device. This effect will be used for high-speed p–n junction modulators) [1].

The change in index of refraction is phenomenologically described by [3]:

$$\Delta n \text{ (at 1550 nm)} = -8.8 \times 10^{-22} \Delta N - 8.5 \times 10^{-18} \Delta P^{0.8}$$

$$\Delta n \text{ (at 1310 nm)} = -6.2 \times 10^{-22} \Delta N - 6 \times 10^{-18} \Delta P^{0.8}$$

and the change in absorption is described by:

$$\Delta \alpha \text{ (at 1550 nm)} = 8.5 \times 10^{-18} \Delta N + 6 \times 10^{-18} \Delta P \text{ [cm}^{-1}\text{]}$$

$$\Delta \alpha \text{ (at 1310 nm)} = 6 \times 10^{-18} \Delta N + 4 \times 10^{-18} \Delta P \text{ [cm}^{-1}\text{]},$$

where N, P are the carrier densities of electrons and holes [cm⁻³].

It should be noted that holes have a smaller absorption as compared with electrons, whereas holes have a larger index shift. Thus, holes are most effective for providing an index shift with minimal absorption, hence modulators typically use holes for offset junction designs* (e.g. in Mach–Zehnder or ring modulators).

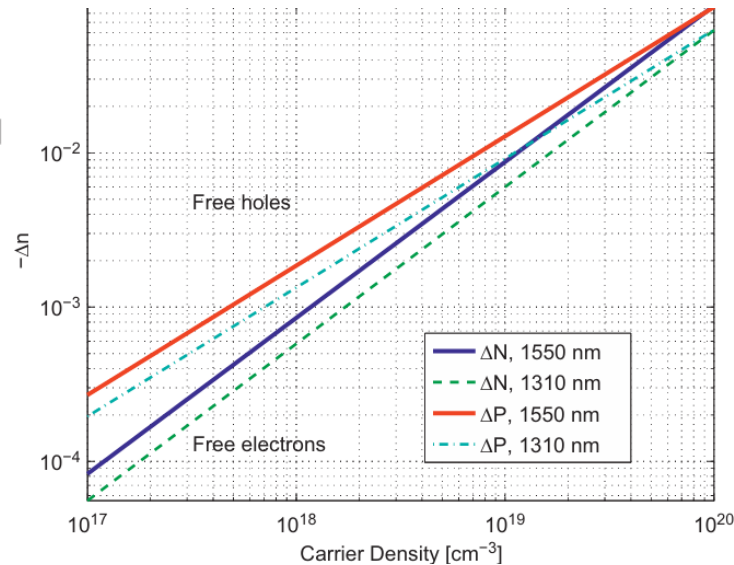
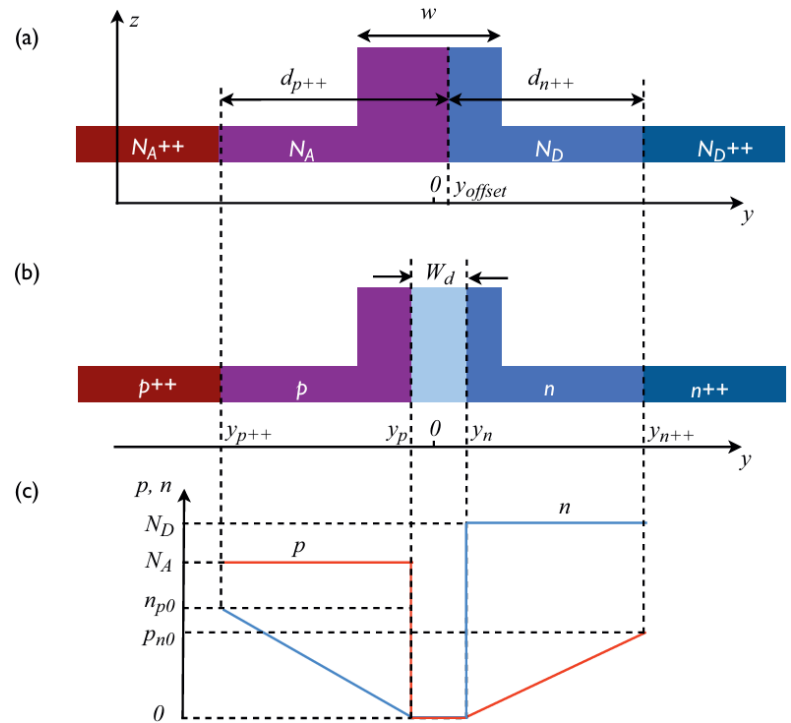


Fig. Change in index of refraction versus carrier density.

* Deliberate spatial displacement introduced in photonic structures—especially in waveguide crossings or couplers—to optimize performance characteristics such as transmission loss and crosstalk.

- The impurity and carrier distributions in a carrier-depletion phase modulator are illustrated in Figure

Fig. p–n Junction in a rib waveguide: (a) cross-sectional view of the impurity distribution assumed in the abrupt junction model; (b) cross-sectional view of the carrier distribution; (c) 1D free-carrier distributions.



The width of the depletion region, W_d , is determined by the impurity densities (N_A and N_D), as well as the applied voltage (V), and is given by [1]:

$$W_d = \sqrt{\frac{2\epsilon_0\epsilon_s(N_A + N_D)(V_{bi} - V)}{qN_A N_D}}$$

where ϵ_s is the relative permittivity and V_{bi} is the built-in or diffusion potential of the junction given by:

$$V_{bi} = \frac{k_B T}{q} \ln \frac{N_A N_D}{n_i^2}$$

- Forward-biasing a p–n junction (or PIN diode) **injects** electrons and holes into the waveguide (increase ΔN) – this raises index and absorption. Reverse-biasing **depletes** carriers (reducing ΔN) – this lowers index (and reduces absorption) [4].

In forward bias, the change in refractive index occurs due to minority carrier injection, and the effect is relatively strong resulting in highly efficient modulation. Although the modulation is very weak below the diode threshold (also referred to as the cut-in or knee voltage). However, just a small voltage beyond threshold injects enough carriers to obtain a π phase shift** [5].

When reverse biased, there is no charge injection and therefore no diffusion capacitance. This leads to a much higher intrinsic bandwidth (f_{3dB} *** of the junction.

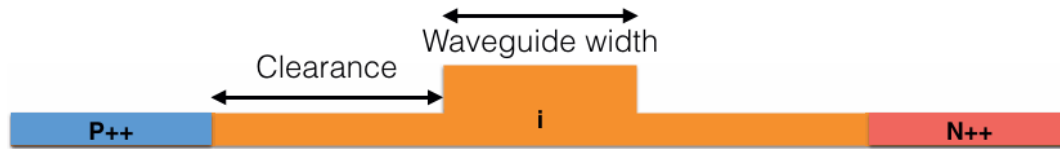
The modulation is performed by the extraction of minority carriers representing the depletion region which overlaps with the optical mode.

**In a MZI, light is split into two paths, when one path accumulates a phase shift of π radians relative to the other, the recombined light waves cancel each other out. This is the basis for on-off keying (OOK) modulation, where binary data is encoded by switching between constructive and destructive interference.

*** It directly impacts the **electro-optic bandwidth** of the modulator. It characterizes how fast the junction can respond to electrical signals for optical modulation. A low f_{3dB} means the junction acts like a bottleneck, limiting how fast the modulator can operate.

The junction capacitance arises from the depletion region, which is also present in the forward biased case but is much smaller in reverse biased diodes. Theoretically, the depletion width increases with increasing reverse bias.

- Similar to the p-n junctions considered above, active silicon photonics devices can be constructed using forward bias PIN junctions. In this case, it is useful to increase the size of the junction by introducing an un-doped (intrinsic) region between the P and N regions, as shown in Figure:



This eliminates the excess optical losses in the waveguide. When the structure is unbiased; thus, the device functions as a conventional rib waveguide. By applying a forward bias, carriers are injected. These carriers change the index of refraction and introduce optical absorption. Hence, this is useful to construct variable optical attenuators (VOA) using the absorption property, and phase tuners using the index of refraction change.

- The carrier concentration for the plasma dispersion effect in silicon can be modulated by various structures. The most commonly studied silicon optical modulator structures are based on carrier accumulation, carrier injection, or carrier depletion. Each structure must be able to guide light while allowing to modulate the carrier concentration in the region of the optical mode [5].

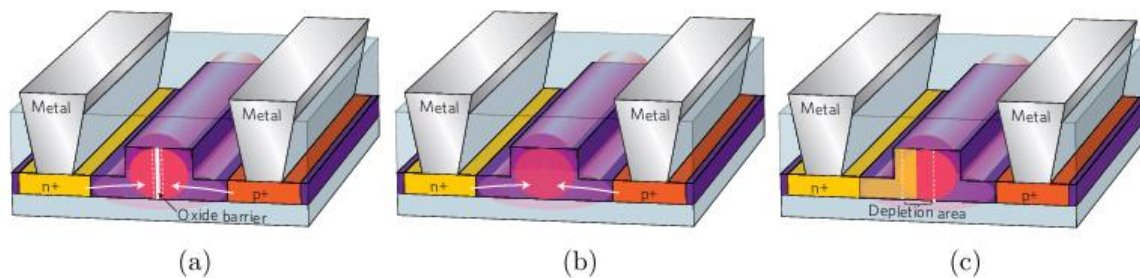


Fig. Methods of plasma dispersion modulation in silicon: (a) carrier accumulation, (b) carrier injection, and (c) carrier depletion.

A carrier accumulation structure is illustrated in fig. (a). In a metal-insulator-semiconductor (MIS) structure with an n-type doped semiconductor, a positive potential applied at the metal would result in holes accumulated at the metal-oxide interface. This accumulation of holes leads to an accumulation of electrons at the oxide-semiconductor interface. The accumulation of charges in the waveguide region modifies the refractive index, thus creating modulation.

Carrier accumulation has a relatively fast response because it involves the movement of the majority carriers and recombination is avoided. MIS structures can typically also operate in depletion and inversion mode.

Carrier injection and carrier depletion modulators use a diode-waveguide structure as illustrated in Fig. (b) and (c) respectively. The mode of operation, either carrier injection or depletion, is determined by the biasing of the diodes.

References

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